



Grover algorithm

Session 3: 6th May 2026





- Known as quantum search algorithm
- Quantum algorithm for unstructured search that finds the unique input to a **black box** function that produces a particular output value.

x : represents a possible input. $x \in \{0, 1\}^n$

There are 2^n inputs

$$f(x) \in \{0, 1\}$$

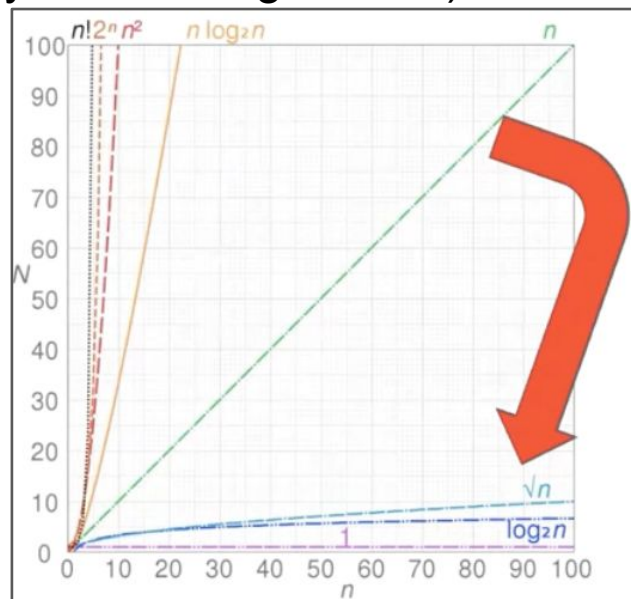
$\exists$ only one ω such that $f(\omega) = 1$, all other $x \neq \omega$, $f(x) = 0$

Hes·so Grover algorithm: The black box function

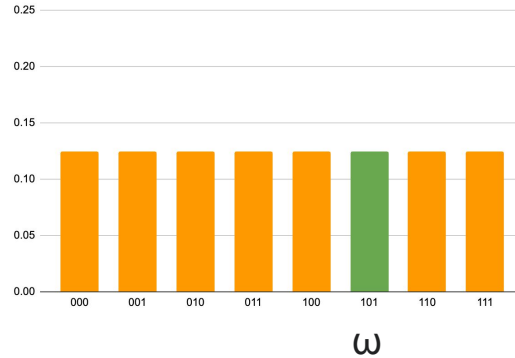
- The f function encodes the problem
- Grover works for problems where:
 - Solution is hard to find (computationally difficult to generate)
 - But easy to verify: calculate $f(x)$
- Complexity
 - Classical computing: $O(2^n)$
 - Quantum computing: $O(2^{n/2})$

Example: $n = 40$

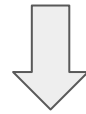
- Classical: $n^{40} \sim 10^{12}$
- Grover: $n^{20} \sim 10^6$



000	0.125
001	0.125
010	0.125
011	0.125
100	0.125
101	0.125
110	0.125
111	0.125



- ω is unknown (hard to find), but ...
- If we calculate $f(\omega)$, the result will be 1: $f(\omega) = 1$

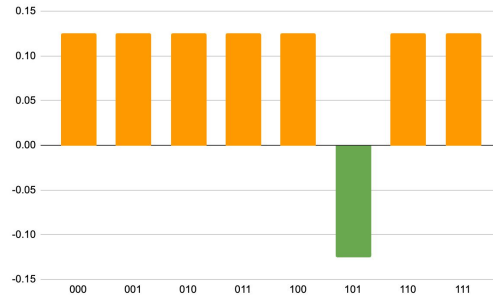


Marking the solution

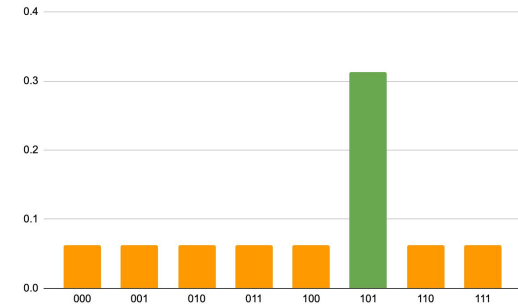
$$a_i = 2 \cdot \text{average} - a_i$$

000	0.0625
001	0.0625
010	0.0625
011	0.0625
100	0.0625
101	0.3125
110	0.0625
111	0.0625

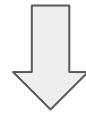
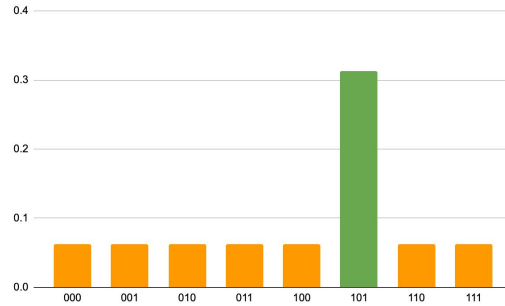
000	0.125
001	0.125
010	0.125
011	0.125
100	0.125
101	-0.125
110	0.125
111	0.125



Amplifying the solution (Diffusion)

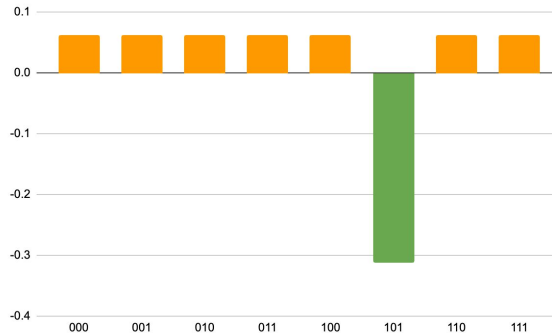


000	0.0625
001	0.0625
010	0.0625
011	0.0625
100	0.0625
101	0.3125
110	0.0625
111	0.0625



Marking the solution

000	0.0625
001	0.0625
010	0.0625
011	0.0625
100	0.0625
101	-0.3125
110	0.0625
111	0.0625

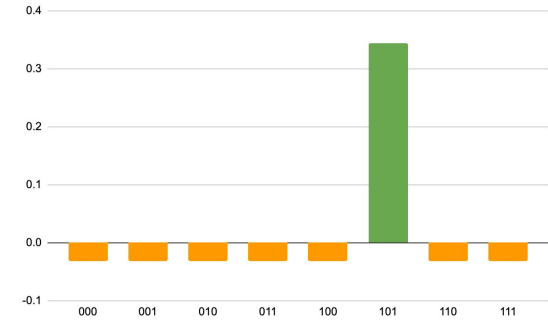


$$a_i = 2 \cdot \text{average} - a_i$$

Amplifying the solution (Diffusion)



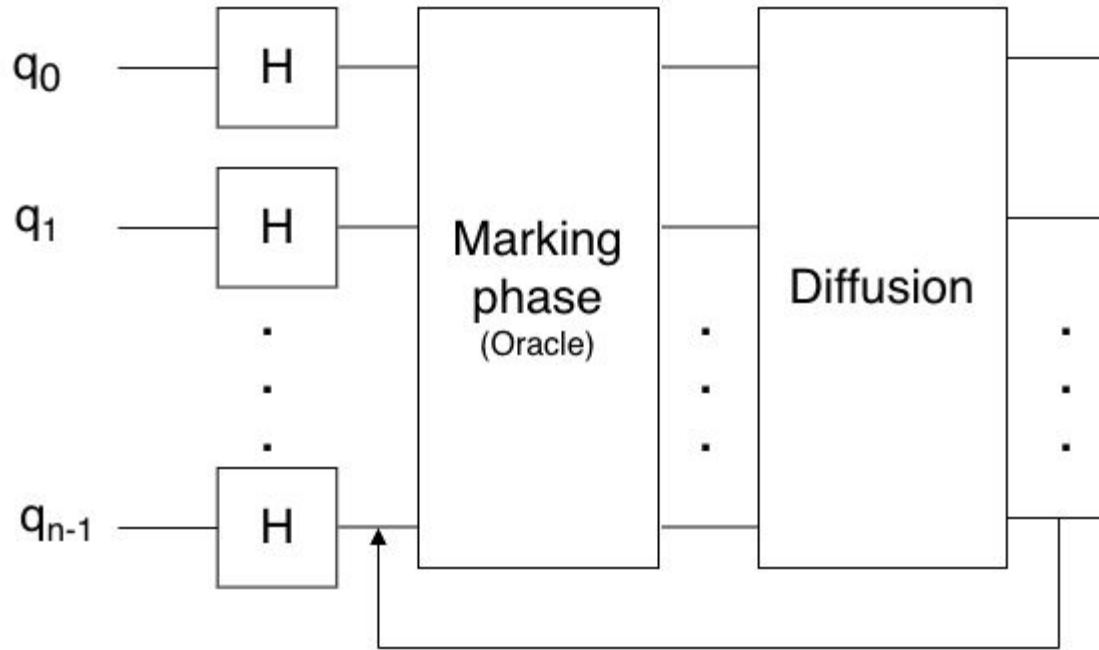
000	-0.03125
001	-0.03125
010	-0.03125
011	-0.03125
100	-0.03125
101	0.34375
110	-0.03125
111	-0.03125





1. Which quantum circuit should be used for the marking phase?
2. Which quantum circuit should be used for the amplifying phase?
3. When to stop the loop ?
4. Why using $a_i = 2 \cdot \text{average} - a_i$ as “amplifier”







- Start by a register of n qubits: $|00000 \dots 00\rangle$.
- $N = 2^n$
- Apply Hadamard : $H^{\otimes n}$ on $|00000 \dots 00\rangle = |\psi_0\rangle$
- $|\psi_0\rangle = \frac{1}{\sqrt{N}} \sum |i\rangle$ with $|i\rangle \in \{|0000 \dots 00\rangle, \dots |11111 \dots 11\rangle\}$





$$O = I - 2 |\omega\rangle \langle \omega|$$

$$O|x\rangle = \begin{cases} |x\rangle & \text{for all } x \neq |\omega\rangle \\ -|x\rangle & \text{for } x = |\omega\rangle \end{cases}$$

$$O|x\rangle = (-1)^{f(x)} |x\rangle$$

Quantum circuit:

1. Apply X to all qubits with state 0 in ω
2. Apply MCZ (phase flip only if all qubits are 111.. 11)
3. Undo the X gates (Apply X to all the qubits to which we applied an X in step 1)

The oracle **marks the solution (ω)** by flipping its sign (phase)





$$D = 2|\psi_0\rangle\langle\psi_0| - I$$

D is the quantum implementation of:
 $a_i \leftarrow 2 \cdot \text{average} - a_i$

$$|\psi_0\rangle = \frac{1}{\sqrt{N}} \sum |i\rangle$$

with $|i\rangle \in \{|0000 \dots 00\rangle, \dots |11111 \dots 11\rangle\}$

The diffusion phase mirrors each amplitude around the average.



$$D = H^{\otimes n} X^{\otimes n} MCZ X^{\otimes n} H^{\otimes n}$$

- The component along $|\psi_0\rangle$ becomes aligned with $|0\rangle$
- This is essentially a change of basis (coordinates)

Apply the phase flip on $|0\rangle$

$$(I - 2|0\rangle\langle 0|)$$

Returns to the original basis



- The first Hadamard transform the basis so that $|\psi_0\rangle$ corresponds to $|0\rangle$
- Middle block: apply the phase flip on $|0\rangle$
- Second Hadamard return to the original basis

